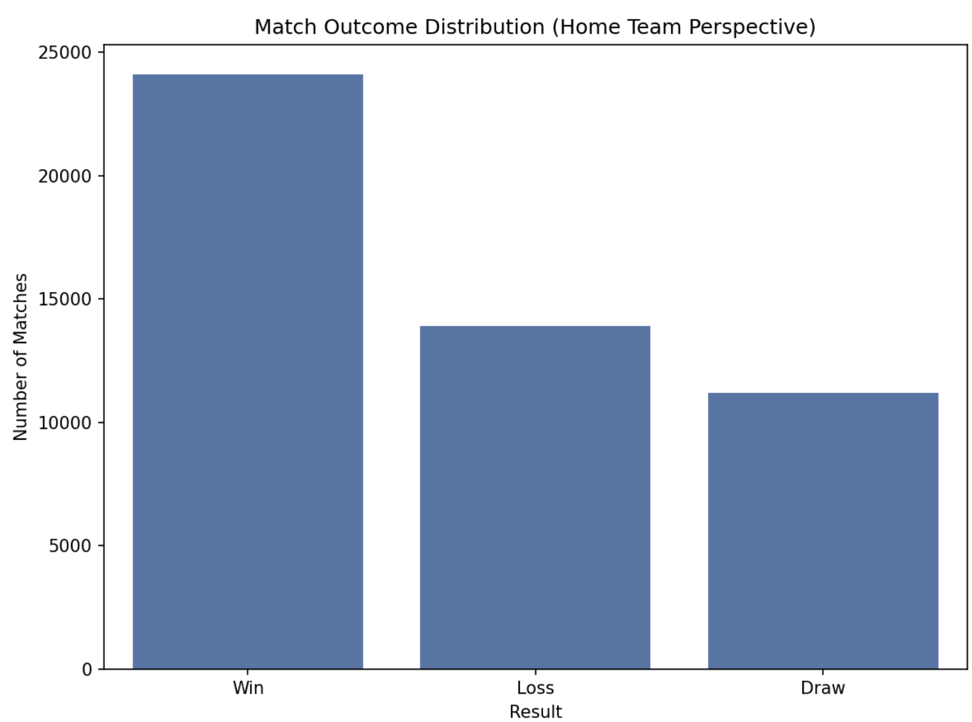
Module 8 Capstone Project:

In today's soccer world, being able to predict match outcomes has become a vital component in matters such as betting markets, scouting, and broadcasting rights. We work with the International Football Results dataset from Kaggle, which contains over 40,000 matches between international teams from 1872 to present day. What we are trying to figure out is if home advantage is a thing and be able to predict either a win, loss, or draw for the home team. To figure this out we use the following techniques: K-Means, Logistic Regression, and Random Forest. K-Means for identifying team archetypes based on scoring behavior, Logistic Regression for interpretable classification, and Random Forest as the more robust ensemble classifier.

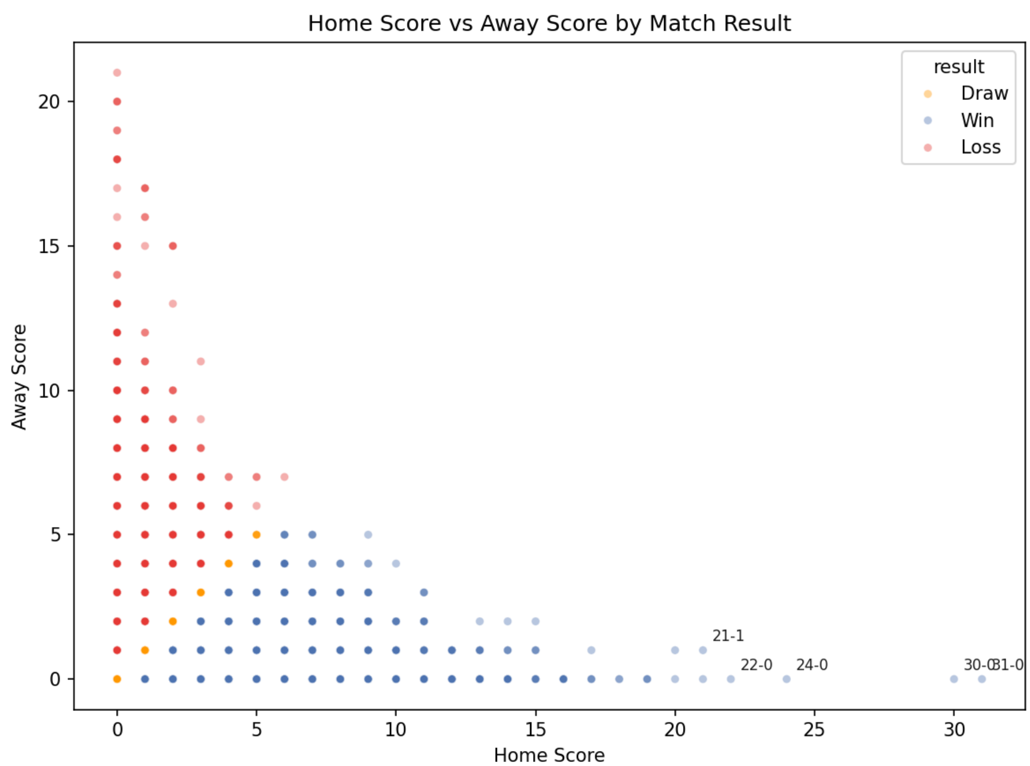
Data Preparation

Before we proceed with some of the techniques, we must investigate the data and prepare it in order to proceed with dissecting what is going on. We'll first start off with the bar chart created to get a general idea.



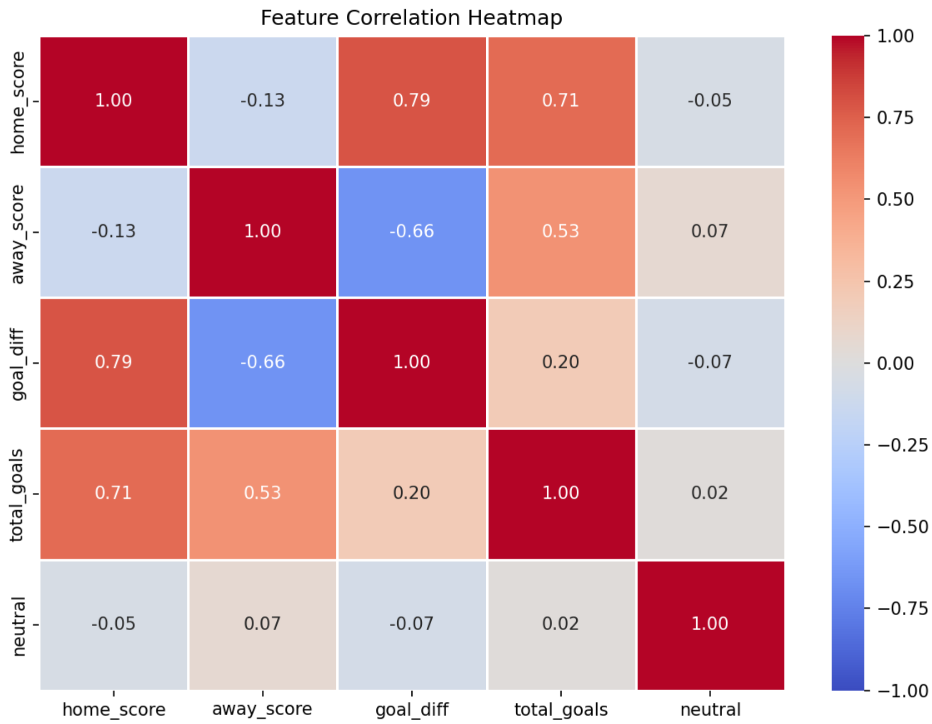
As we look at the bar chart, we can see the distribution of match outcomes across the dataset. As expected, teams who play at home win more often than they lose or draw, which reflects the often-understood phenomenon of home advantage in soccer. There are a variety of factors that contribute to this effect. Factors such as crowd support and familiarity with the home pitch are common. Other factors such as less travel fatigue compared to the away team and changes in climate and altitude, because it is international soccer, are less talked about in mainstream media and are often overlooked but play a vital role in the uncalculated factors when it comes to predicting results.

In addition to the bar chart, we also included a scatter plot to help us map home scores against away scores, which are colored by the result of the match.



As labeled, yellow represents a draw, blue for a win, and red for a loss. The axes map out the results respectively for the home team and away team, with a few labeled scorelines included for reference. Looking at the labeled points, we can see that there are scores that are extreme outliers, with 31-0 and 30-0 being the standouts. This is particularly because of the early years' entries where the competition between nations was highly unequal. In addition, documentation of early scores and data collection is often disputed. For example, Pelé, considered one of the greatest soccer players ever, has had his actual goal scoring tally put into question, where it is claimed to be around the thousands, while sources note that some of those goals came from unofficial matches that otherwise would not have been counted. This makes those earlier data entries less reliable than the more present-day entries. Despite being technically valid entries, these data points pose potential risks in distorting the model by teaching it patterns that would almost never occur in today's game compared to the level of documentation back then. The last visual we included before implementing the techniques is the heatmap, where we look at how well we can predict results based on the features listed:

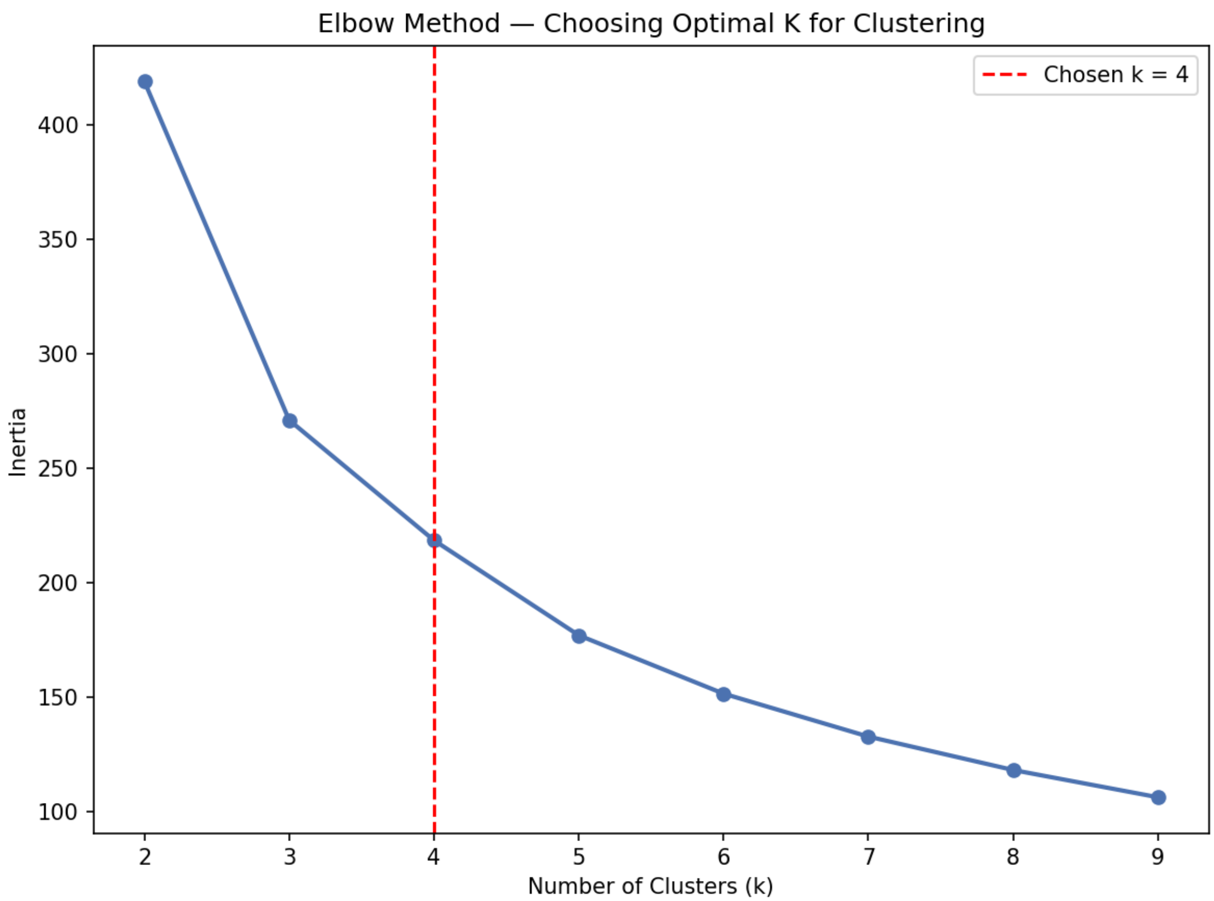
- home_score: the score (number of goals scored) by the home team.
- away_score: the score (number of goals scored) by the away team.
- goal_diff: the goal difference between the home and away teams, depending on who won or lost, one would have a positive difference if they won and the other will have a negative score if they lost. If a draw, the difference will be zero.
    - For example, if the score was 3-1 to the home team, the home team will have a goal difference of +2 while the away team will have a goal difference of -2.
- total_goals: the total goals scored in the game:
- neutral: this is when the game is not played at home ground, instead played at a neutral venue.



Before we look at the heatmap, I stumbled upon the discovery when preparing the data of a leakage in the feature selection. As you can tell from the explanation of each feature and looking at the heatmap, it is clear why that is the case. The map reveals the correlation between away scores and home scores, positive and negative. Because goal difference is computed directly from those same scores used to define the Win, Draw, or Loss target variable, the model was essentially given the answer before making any prediction. As a result, the models produced 100% accuracy, making it not a meaningful result. I then corrected this by using total goals and neutral as the sole features, to ensure that the model had to learn from independent signals rather than being given the answers.

Technique 1: K-Means Clustering

The first technique used was K-Means Clustering, to create team-level profiles to organize the teams rather than by individual matches. Teams were categorized by averaged goals scored at home matches, averaged goals conceded at home matches, and overall win rate at home. Teams with fewer than ten home matches were excluded to ensure these profiles were based on sufficient data. This way we can gauge an understanding of a team's playing style and connect the dots to some teams in today's game. In order to determine the optimal number of clusters, the use of the elbow method was needed for this situation.

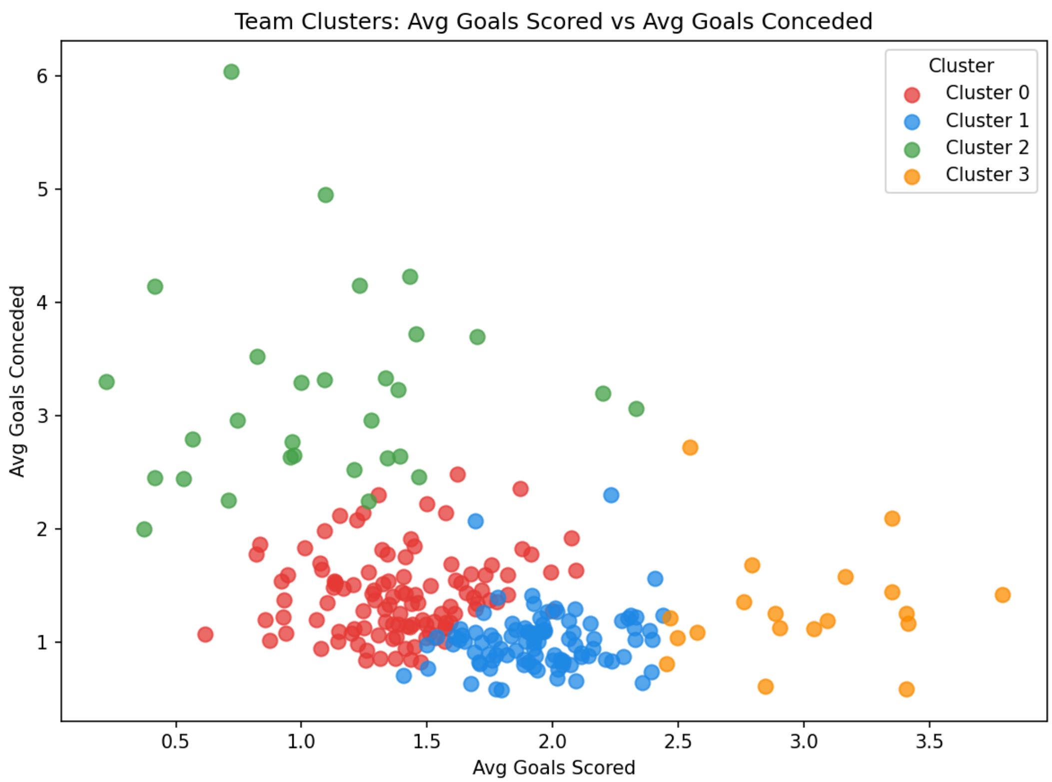


Looking at the visualization, the inertia drops rapidly between k=2 and k=4. After k=4, the plot begins to slow down and plateau, making four the best choice for the number of clusters. Given the number of clusters to look at, we take a look at the K-Means model and each profile:

Cluster Profiles:

cluster	avg_scored	avg_conceded	win_rate
0	1.392	1.414	0.380
1	1.951	1.020	0.557
2	1.087	3.190	0.218
3	2.987	1.306	0.619

As we can see from the terminal output, Cluster 3 represents the strongest of the four groups, averaging 2.99 goals per game and a win rate of 62%. Cluster 2 acts as the representation for the weakest teams, averaging 3.19 goals conceded per game and only a win rate of 22%. Clusters 0 and 1 are the average teams, where one group scores a bit more and concedes a bit less, and vice versa for the other



The visualization sheds light on the numbers, and we can better see the nature of each group. Cluster 1, in blue, while slightly average, is stronger and sits closer to Cluster 3 in orange as one of the more dominant groups in terms of win rate and goal scoring. Cluster 0, in red, is not too far off from Cluster 1 but is not as proficient in scoring and is slightly more susceptible to conceding. Nowhere near as comparable to Cluster 2, in green, which is the weakest group due to its trend in conceding and low scoring averages. For comparison, Cluster 0 would represent nations in North America, Asia, or mid-tier European nations who are often competitive enough. Cluster 1 would represent nations like Italy, good enough to score well and defensively strong. Cluster 2 would represent nations that are still developing their soccer programs and are often much weaker compared to the rest. Cluster 3 would represent nations like Brazil and Spain, renowned for their ability to break down defenses and score frequently, while being more prone to conceding than Cluster 1. While it is hard to pin down which teams from which eras, this distribution captures what you would expect from the FIFA world rankings and general knowledge of certain teams and their play styles, some remaining consistent, others evolving and fluctuating between clusters over time.

Technique 2: Logistic Regression

The second technique used was Logistic Regression, where the model is used to predict categorical outcomes. The model was trained to predict match results using two independent features — since the original choice was flawed and caused a leak — those being total goals and neutral venue. A pipeline was built to handle imputation, feature scaling, and model fitting in a single workflow. The dataset was split into 60% for training and 40% for testing. The Logistic Regression model managed to achieve an accuracy of 47%. Below is the confusion matrix along with the performance metrics:

Logistic Regression Results:

Confusion Matrix:

[[3537 247 698]

[2225 1046 2320]

[3541 1397 4675]]

Accuracy: 0.4703

Precision: 0.4937

Recall: 0.4703

F1-score: 0.4524

The model has a difficult time with draw predictions, which is not surprising as draws are often difficult to predict in soccer. A 0-0 and a 1-1 result are similar in goal counts but result in very different contextual dynamics. Considering that predicting a win, loss, or draw gives a random likelihood of about 33% for each outcome; the model achieving 47% accuracy is genuinely better than chance.

Technique 3: Random Forest

The third technique used was the Random Forest Classifier, which is more robust than the Logistic Regression model. The same pipeline structure and train-test split were used to maintain consistency. The Random Forest model was able to achieve an accuracy of 51%. Below is the confusion matrix, performance metrics, and feature importance chart:

Random Forest Results:

Confusion Matrix:

[[4297   43 142]

[1535 2229 1827]

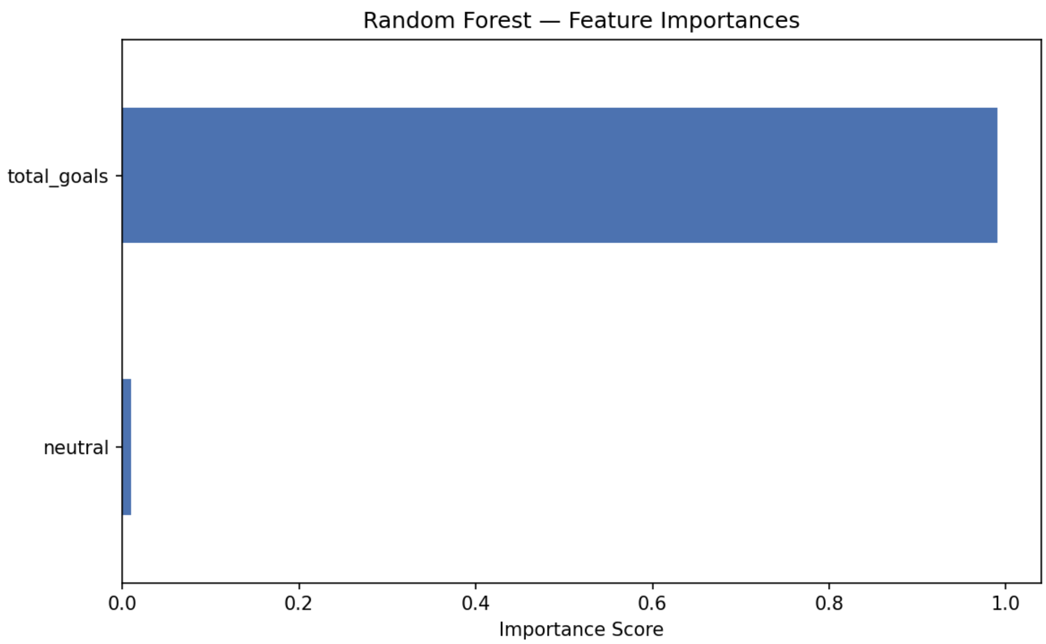
[2807 3169 3637]]

Accuracy: 0.5163

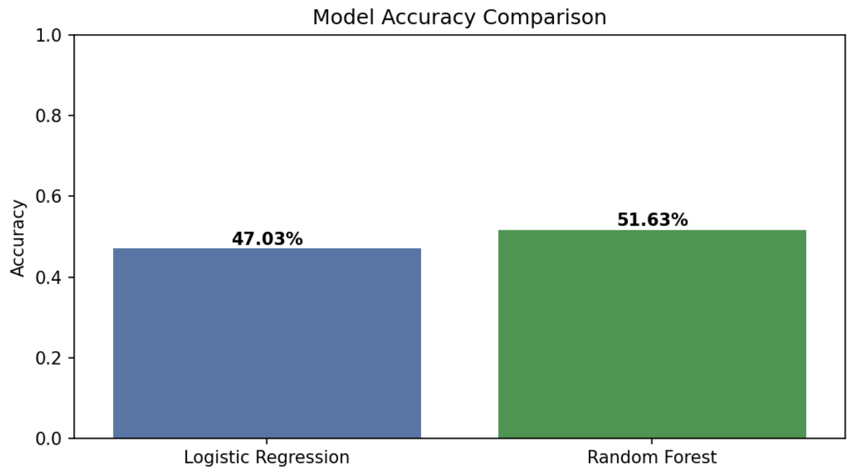
Precision: 0.5464

Recall: 0.5163

F1-score: 0.4973



The visualization shows the importance of each feature, depicting which will be most influential in helping the model. We then compare the two models, Logistic Regression and Random Forest, side by side.



This project proved to be important in gaining insight into international soccer while presenting the challenges of predicting match outcomes using data mining techniques. The techniques and visualizations helped confirm a few already understood concepts while also uncovering interesting findings about past data points and why some numbers and results are what they are. The discovery of the flawed feature selection was an interesting find that made complete sense once identified; the choice of certain features can dictate the effectiveness of a model's predictions entirely. There are a few limitations worth pointing out, from the wide timespan of the dataset to the fact that the dataset only records outcomes rather than pre-match conditions. Future work incorporating pre-match features such as FIFA world rankings, recent form, and player availability would be needed to build a more meaningful and deployable prediction tool. In conclusion, this project served as a great learning experience, combining everything learned over the course of the class and applying these techniques to a real-world problem.